

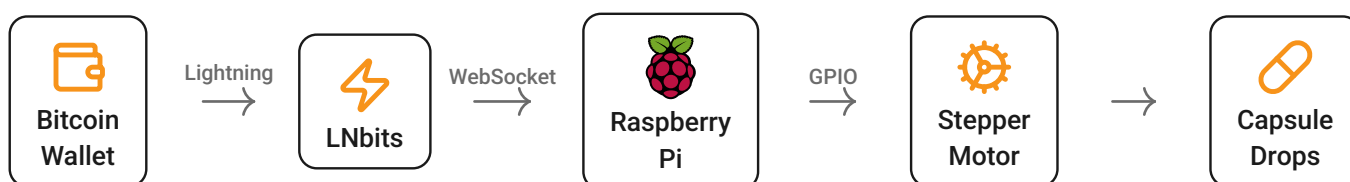


LIGHTNING GACHA



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Scan the QR, send a **Lightning payment**, and a capsule drops out.



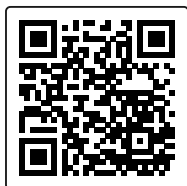
HOW IT WORKS

- ▶ **Bitcoin Wallet** — any Lightning wallet on your phone (Phoenix, Wallet of Satoshi, Zeus...).
- ▶ **LNbits** — open-source Lightning payment server with a WebSocket API.
- ▶ **Raspberry Pi** — subscribes to payment events and drives the motor.
- ▶ **Stepper Motor** — rotates a paddle inside the gacha to release one capsule per turn.

HARDWARE

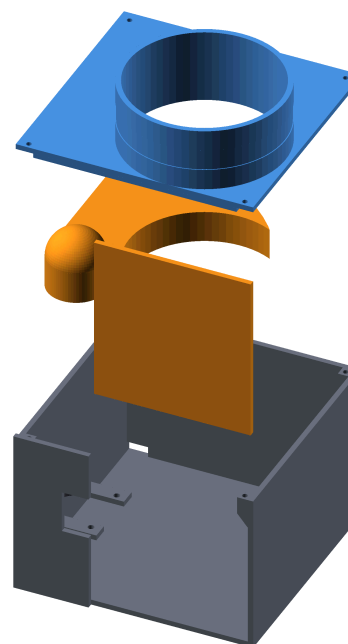
- ▶ **Body** — 3D-printed in three parts, modeled in OpenSCAD.
- ▶ **Dispenser** — $\varnothing 75$ mm capsule chamber with a stepper-driven paddle.
- ▶ **Controller** — Raspberry Pi Zero 2.
- ▶ **Motor** — NEMA-17 17HS4401, 1.8°/step, 42 N·cm.
- ▶ **Motor Driver** — TMC2209 (silent 1/8 microstepping → 1600 steps/rev).
- ▶ **Power** — USB-C PD trigger (12 V) → LM2596 step-down.

OPEN SOURCE



All 3D models and software are on GitHub.

<https://github.com/aostanin/jrrf-gacha>



Printed Parts